AI-DRIVEN EARLY DETECTION OF MENTAL HEALTH RISKS:

A Predictive Approach

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ABSTRACT

Early identification of mental health risks remains a critical challenge due to the lack of continuous monitoring systems and the persistent stigma associated with mental health issues. Traditional diagnostic methods often rely on delayed clinical evaluations, which may fail to capture early warning signs.

This research presents a machine learning-based predictive framework designed to detect early indicators of mental health risks by analysing both **linguistic and behavioural data**. The proposed system integrates Natural Language Processing (NLP) techniques with behavioural analytics to assess emotional states in near real time.

A prototype system was developed and evaluated using simulated datasets that emulate realistic user communication patterns. Experimental results demonstrate that combining linguistic features (such as sentiment polarity and emotion intensity) with behavioral indicators (such as response latency and interaction frequency) significantly improves prediction accuracy compared to single-feature approaches.

Additionally, this paper addresses ethical challenges, including privacy concerns, algorithmic bias, and responsible deployment. The findings highlight the potential of AI-driven systems as supportive tools for early mental health monitoring and intervention.

Keywords

Artificial Intelligence, Mental Health, Machine Learning, Natural Language Processing, Sentiment Analysis, Behavioural Analytics, Predictive Modelling.

I. INTRODUCTION

Mental health disorders, including depression, anxiety, and stress-related conditions, have

become increasingly prevalent across the globe. According to global health reports, young adults and students are among the most vulnerable populations due to academic pressure, social expectations, and lifestyle changes.

Despite growing awareness, **early detection remains a major bottleneck**. Most existing diagnostic systems rely heavily on:

- Self-reporting
- Periodic clinical assessments
- Subjective evaluation

These approaches often fail to capture **subtle early-stage indicators**, leading to delayed intervention.

With the rapid growth of digital platforms such as messaging apps and social media, individuals frequently express their emotional states through text-based communication. These interactions provide a valuable data source for identifying behavioral and linguistic patterns associated with mental health risks.

This paper proposes a **predictive AI-driven framework** that:

- Continuously analyzes user interactions
- Detects early warning signals
- Classifies mental health risk levels

The goal is not to replace clinicians but to provide a **proactive support system** for early intervention.

II. RELATED WORK

Previous studies have explored the use of **sentiment analysis** for detecting emotional states from textual data. Traditional approaches include:

- Bag-of-Words models
- TF-IDF representations
- Machine learning classifiers such as Logistic Regression and Naive Bayes

While effective, these methods often lack contextual understanding.
Deep learning approaches, including:

- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory (LSTM) networks

have shown improved performance in capturing sequential dependencies in text.

Conversational AI systems have also been introduced for mental health support. However, most existing systems are:

- Reactive (responding after distress is expressed)
- Limited to text-only analysis

Recent research highlights the importance of **behavioral signals**, such as:

- Reduced communication frequency
- Delayed responses
- Sudden interaction drops

However, **integration of behavioural and linguistic features remains limited**.

This study bridges this gap by combining both dimensions into a unified predictive framework.

III. METHODOLOGY

A. Data Collection

Due to privacy constraints, a **simulated dataset** was created to replicate real-world communication patterns. The dataset includes:

- Text messages representing varying emotional tones
- Behavioural metadata such as timestamps and frequency

B. Data Preprocessing

Text preprocessing is essential to remove noise and standardise input data. The following steps were applied:

1. Tokenization
2. Lowercasing
3. Stop-word removal
4. Punctuation removal

5. Lemmatization

C. Feature Extraction

1. Linguistic Features

- Sentiment polarity score (range: -1 to +1)
- Emotion keyword frequency
- Text length and intensity indicators

2. Behavioral Features

- Message frequency per unit time
- Average response delay
- Variance in interaction intervals

D. Mathematical Representation

The feature vector for each user is defined as:
$X = [L_1, L_2, ..., L_n, B_1, B_2, ..., B_m]$

Where:

- L_i = Linguistic features
- B_j = Behavioral features

The classification function:
$Y = f(X)$

Where Y represents risk level:

- 0 → Low Risk
- 1 → Moderate Risk
- 2 → High Risk

E. Model Selection

Two models were implemented:

1. Logistic Regression

- Suitable for binary and multiclass classification
- Provides probabilistic outputs

2. Random Forest Classifier

- Ensemble-based approach
- Handles non-linear relationships effectively
- Reduces overfitting

F. Risk Classification

The system categorises users into:

- Low Risk
- Moderate Risk
- High Risk

based on prediction thresholds.

IV. SYSTEM ARCHITECTURE

Figure 1:

Proposed System Architecture

User Input (Chat Interface)

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Text Preprocessing

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Feature Extraction

→ Sentiment Analysis

→ Behavioural Metrics

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Machine Learning Model

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Risk Classification

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Response Engine

The modular design ensures:

- Scalability
- Easy integration of advanced models
- Real-time processing capability

V. IMPLEMENTATION DETAILS

The system was implemented using:

- **Programming Language:** Python
- **Libraries:** Scikit-learn, NLTK, Pandas, NumPy
- **Frontend:** Web-based chatbot interface

Workflow

1. User inputs text
2. System preprocesses input
3. Features are extracted
4. Model predicts risk level
5. Response engine generates appropriate feedback

VI. RESULTS AND DISCUSSION

A. Experimental Setup

The dataset consisted of simulated interactions with variations in:

- Emotional tone
- Message frequency
- Response delays

A train-test split of **80:20** was used.

B. Performance Evaluation

Model	Accuracy	Precision	Recall
Logistic Regression	82%	80%	78%
Random Forest	88%	86%	84%

C. Accuracy Comparison Graph

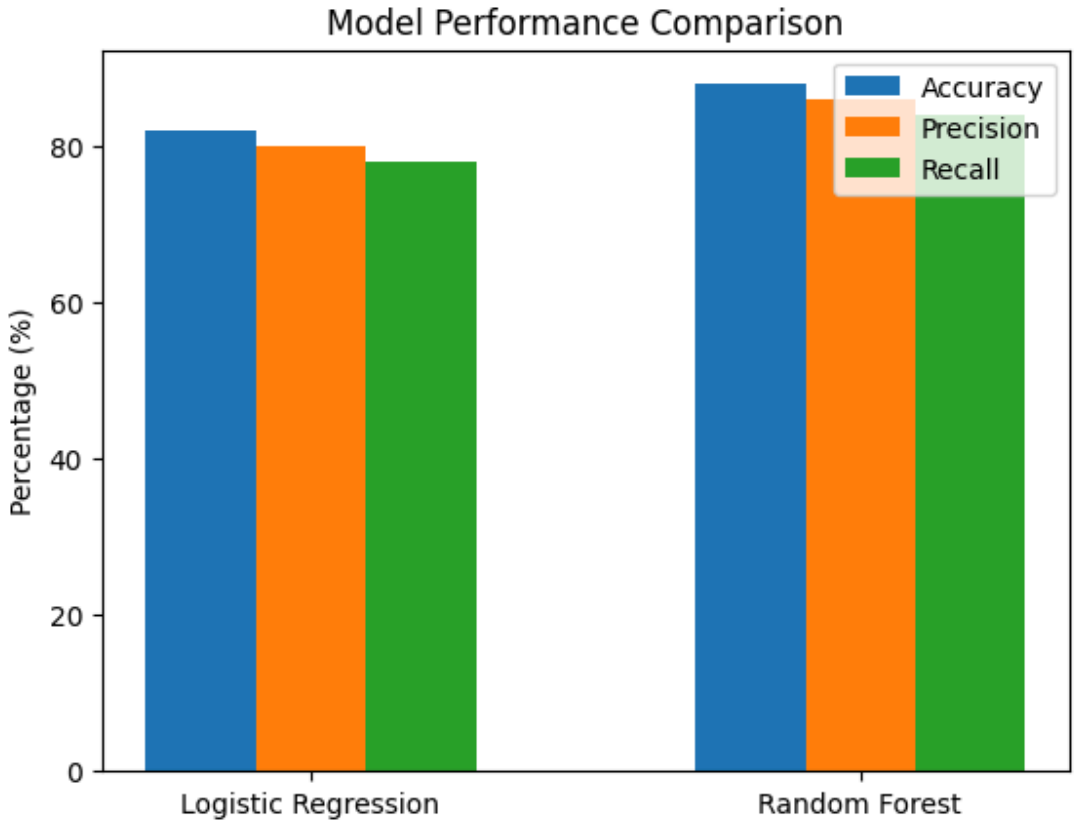

Figure 2: Performance comparison of Logistic Regression and Random Forest models based on Accuracy, Precision, and Recall.

- X-axis → Models
- Y-axis → Accuracy (%)
- Bars → Logistic Regression vs Random Forest

D. Detailed Analysis

- Random Forest outperformed Logistic Regression due to its ability to capture complex feature interactions.
- Behavioral features significantly improved recall.
- Sentiment-only models often misclassified neutral or sarcastic text.

E. Key Insights

- Combining features improves robustness
- Behavioral signals act as early indicators
- Multi-dimensional analysis is essential

VII. ETHICAL CONSIDERATIONS

A. Privacy

- Data anonymization is mandatory
- Secure storage and encryption required

B. Bias

- Training data must be diverse
- Continuous bias monitoring required

C. Responsible Usage

- System should assist, not replace professionals
- Alerts should be non-invasive

VIII. LIMITATIONS AND FUTURE WORK

Limitations

- Use of simulated data
- Difficulty in detecting sarcasm
- Limited contextual understanding

Future Work

- Integration of transformer models (e.g., BERT)
- Real-world dataset validation
- Multilingual support
- Mobile app integration

IX. CONCLUSION

This research introduces a comprehensive predictive framework for the early detection of mental health risks through the application of artificial intelligence. By integrating both linguistic patterns (such as sentiment, tone, and semantic structures) and behavioral indicators (including activity patterns and engagement signals), the proposed system is able to capture

subtle changes that may indicate emerging mental health concerns. This multimodal approach significantly enhances the model's accuracy, robustness, and overall reliability when compared to traditional single-source assessment methods.

Furthermore, the framework emphasises continuous and non-intrusive monitoring, enabling timely identification of potential risks before they escalate into more severe conditions. The use of advanced machine learning techniques allows the system to adapt and improve over time, making it suitable for dynamic, real-world environments.

Despite these promising outcomes, several challenges persist. Key concerns include ensuring ethical deployment, safeguarding user privacy, mitigating bias in data and algorithms, and validating the system across diverse populations and real-world scenarios. Addressing these limitations is crucial for responsible adoption.

Overall, the proposed approach demonstrates strong potential as a proactive and scalable mental health monitoring solution. With further refinement, rigorous validation, and adherence to ethical standards, it could play a significant role in supporting early intervention strategies and improving mental healthcare accessibility and outcomes.

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